

Title: Metagenomics tools & approaches: revealing the hidden diversity in Agricultural fields

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1. Introduction:

This exercise has been performed as part 2 of the APS Plant Health 2026 pre-meeting workshop on “Genomics and Metagenomics tools and approaches: revealing hidden diversity in agricultural fields”. Part 1 (DOI of part A paper) of the Plant Health Teaching module focused on genomics and the extension of variant-calling methods in genomic samples. Samples collected from field environments or natural ecosystems are rarely composed of a single genome. Instead, they often contain complex mixtures of closely related strains that coexist, interact, and evolve simultaneously, giving rise to multiple lineages within the same habitat. Characterizing genetic variation in these heterogeneous populations therefore requires metagenomic approaches that capture diversity across entire communities and extend beyond traditional isolate genome analyses.

Standard metagenomic workflows typically start with community-level profiling tools such as Kraken2 (Wood et al. 2019), Centrifuge (Kim et al. 2016), or Bracken (Lu et al. 2017). These methods are based on K-mer-based approaches or marker gene alignment to classify sequencing reads across broad taxonomic profiles at domain, phylum, genus and species levels. Community-profiling tools are highly efficient at detecting overall community or microbiome composition in metagenomic samples but lack the sensitivity to distinguish closely related strains. Although some tools are sensitive at the strain level, such as panphlan and metaMLST (Scholz et al. 2016; Zolfo et al. 2017), these tools rely on the assumption that a single dominant strain is present in each species, whereas agricultural samples may contain closely related strains. This is exactly what we found in our recent study. Using strain-resolved metagenomics to identify pathogen diversity, we found multiple closely related strains of a single pathogen species at varying abundance levels within a single field (Bhandari et al. 2025). This finding highlights the importance of strain-level identification and profiling for high-resolution epidemiology and outbreak tracking. The StrainEst package was used for strain profiling of the metagenomic samples in that study.

Tools like StrainEst, inStrain, and MIDAS utilize comprehensive genome SNV profiling across entire reference genomes. Among these, inStrain outperforms tools such as StrainPhlan and MIDAS for assessing microdiversity and SNV comparisons, but it cannot efficiently detect differences beyond 99% ANI (Truong et al. 2017; Nayfach et al. 2016; Olm et al. 2021). StrainEst uses regression-based models to deconvolve the read set into precise strain abundances across multiple strains (Albanese and Donati 2017).

The focus of this exercise will be on strain-level identification using StrainEst. This tool utilizes SNP profiles from reference genomes to quantitatively estimate strain abundance in metagenomic or mixed-genomics samples (Albanese and Donati 2017). This SNP-based approach gives this tool high accuracy in distinguishing closely related strains in environmental samples. StrainEst is particularly useful for epidemiological surveillance, outbreak investigations, and the study of population dynamics within hosts or environmental niches. Researchers can determine which strains are present in a sample, estimate their relative abundances, and investigate how strain populations change over time or across different geographic locations. The workflow uses high-resolution SNP markers derived from reference-strain diversity, together with regression-based statistical models, to separate mixed samples, assign sequencing reads to specific strains, and accurately estimate strain abundance.

In this exercise, we will use *Xanthomonas*-infected field samples to demonstrate both inter- and intra-species identification.

2. Primary Goal:

Extend the concepts learned in exercise 1 to complex metagenomic datasets to characterize strain-level diversity within mixed bacterial populations

3. Learning Objectives:

By the end of this workshop, participants will learn the following:

- Construction of lineage-specific reference signatures by building single-nucleotide variant (SNV) signature matrices and sequence cluster alignments from curated bacterial reference genomes to establish multi-strain database scaffolds.
- Mapping the paired-end metagenomic sequencing reads against multi-strain reference alignments.
- Application of Lasso-based regression models in the *StrainEst* pipeline to distinguish true biological strain variations from sequencing artifacts and determine both the identity and relative abundance of coexisting field strains.
- Critically analyze strain shifts across samples, identify depth biases, and diagnose false positives or false negatives due to missing reference strains in public repositories.

4. Duration:

This exercise is designed to be completed in about 2 hours.

5. Setting Materials:

- Laptop with internet access and 16- 20 GB of RAM.
- Terminal with R/RStudio and Python installed. If not available, web tools will be provided as alternatives.
- A spreadsheet or text editor.

6. Material & Software:

To perform this exercise, researchers need the modules/tools/packages mentioned below in their High-Performance Supercomputing (HPC) system profiles or on an offline Docker system.

- **Software/Tools:** Strainest (<https://github.com/compmetagen/strainest>), Bowtie2, samtools
- **Genomic data:** Metagenomics samples, reference genomes of focal species lineages

7. Workshop Protocol:

General Strategy:

Analyzing the field metagenomics samples follows the following hierarchical workflows:

1. Broad microbial community screening: Fast, k-mer-based tools (such as Kraken2, Centrifuge, or even MIDAS) are used to first profile the microbial community composition and confirm the presence and relative abundance of the target pathogen genus (here, *Xanthomonas*).
2. Inter-Species Profiling: Once the target genus is confirmed, map the reads to a multi-species reference database (here, *X. perforans*, *X. euvesicatoria*, *X. vesicatoria*, *X. gardneri*) to determine which species is dominant in the sample or whether co-infection is present.
3. Intra-Species Profiling: A high-resolution, single-species database (here, *X. perforans* lineages) is built to deconvolve closely related, coexisting strains (>99% identity) and to calculate their exact relative proportions.

The last two parts have two subsections (Inter-species and Intra-species diversity estimation), each having similar two phases:

- **Reference Database Construction:** Curate a database of reference genomes representing known species lineages. Extract SNPs that distinguish among references and group references into clusters (lineages) based on SNP similarity. This creates a lineage-specific SNP signature database.
- **Metagenomic Sample Analysis:** Map metagenomic reads to the reference database using fast read alignment (Bowtie2) (Langmead and Salzberg 2012). Then, convert to BAM (binary alignment format), and sorting is performed using samtools (Li et al. 2009). Quantify read coverage at SNP sites. Apply Lasso-based regression to estimate the set of strains/lineages needed to explain observed read patterns and their relative abundances.

Protocol Steps:

A. Inter-species diversity

Step 1: Inter-species reference database construction

In a metagenomics environmental sample, we initially do not know which taxa or species are present. Therefore, it is essential to first identify the species in the samples before examining intra-species diversity or analyzing closely related strains. To start this building, the comprehensive reference database enables detection of unexpected taxa (e.g., contamination, mixed infections) in samples and provides a hierarchical

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scaffold for subsequent intra-species analysis. By first defining which species and strains are present, you establish a realistic search space, reduce computational burden, and improve the accuracy of downstream strain abundance estimation.

Note 1: Here, we outline the following steps for HPC systems that already have the required packages installed. Therefore, we load the module directly rather than installing packages. If a student is using an HPC system, they should check with their service to confirm that the required packages are downloaded. Otherwise, they have to install those packages by following the instructions mentioned on the GitHub pages of the required packages.

Note 2: We are including the HPC-based command lines to make it easier for users to understand the purpose of each command. We have a separate section for Windows and Mac using the offline system (Docker) because the HPC systems are unavailable for the workshop purpose; all the commands are available on our GitHub (<https://github.com/Potnislalab/APS-Plant-Health-Workshop-2026>).

Below are the steps with descriptions:

Command using High-Computing Performance Systems	Description
<code>#!/bin/bash</code> <code>module load strainest</code> <code>strainest mapgenomes Xp91-118.fna Xeu_85-10.fna</code> <code>Xg_USA01-FM3.fna Xv_LMG911.fna Xp91-118.fna MR.fna</code>	# Load the strainest module # Map specific species genomes to the reference genome (Xp91-118.fna) to create a multiple sequence alignment file (MR.fna).
<code>strainest map2snp Xp91-118.fna MR.fna snp.dgrp</code>	# Extract SNPs from the multiple alignment (MR.fna) against the reference, generating a SNP profile file (snp.dgrp).
<code>strainest snpdist snp.dgrp snp_dist.txt hist.pdf</code>	# Calculate distances based on SNP profiles, outputting a distance matrix and histogram.
<code>strainest snpclust snp.dgrp snp_dist.txt snp_clust.dgrp</code> <code>clusters.txt</code>	# Cluster the genomes based on SNP distances to group similarities.
<code>strainest mapgenomes Xp91-118.fna Xeu_85-10.fna</code> <code>Xg_USA01-FM3.fna Xv_LMG911.fna Xp91-118.fna MA.fna</code>	# Generate a final representative multiple sequence alignment (MA.fna) for mapping.
<code>module load bowtie2/2.5.1</code> <code>source /apps/profiles/modules_asax.sh.dyn</code> <code>module load bowtie2/2.5.1</code>	# Setup environment for Bowtie2 (Note: redundant loading from the original script is kept for fidelity)
<code>bowtie2-build -f MA.fna MA</code>	# Build a Bowtie2 index from the multiple alignment FASTA file (MA.fna).

Step 2: Read Mapping and quantification

Command	Description
<code>#!/bin/bash</code> <code>source /opt/asn/etc/asn-bash-profiles-special/modules.sh</code> <code>module load bowtie2/2.5.1</code> <code>module load samtools/1.11</code> <code>module load strainest</code>	# Source the bash profile to initialize the module system on the cluster. # Load the required bioinformatics tools and their specific versions into the environment.

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bowtie2 --very-fast --no-unal -x /path/to/MA -1 sample_1.fq -2 Sample_2.fq -S Sample.sam	# This command currently runs on a single pair of 'fake_sample' files. # Map the paired-end reads to the Bowtie2 index we created. # --very-fast : Uses a faster, slightly less sensitive alignment preset. # --no-unal : Suppresses SAM records for reads that failed to align. # -x : Specifies the path to the Bowtie2 index prefix (MA). # -1 / -2 : Specifies the forward (trim1) and reverse (trim2) paired-end read files. # -S : Specifies the output SAM file name.
samtools view -b Sample.sam > Sample.bam	# Convert the human-readable SAM file into a compressed binary BAM file. # '-b' tells samtools to output in BAM format.
samtools sort Sample.bam -o Sample.sorted.bam	# Sort the BAM file by genomic coordinates. This is required by downstream tools. # '-o' specifies the output sorted BAM file.
samtools index Sample.sorted.bam	# Create an index (.bai) for the sorted BAM file to allow fast lookups.
strainest est path/to/snp_clust.dgrp Sample.sorted.bam \ ./Sample	# Run the final strain estimation using the sorted BAM file and the clustered SNP profile. # It outputs the estimated strain abundances into the './fake_sample' directory.

For each sample, the pipeline generated an output directory containing files that summarize the results. The **abund.txt** file reports the predicted abundance of each reference genome, **max_ident.txt** shows the percentage of alleles from each reference genome detected in the metagenomic sample, **info.txt** provides summary statistics for the prediction (including the Pearson correlation coefficient), **counts.txt** contains the allele counts at each SNV position, and **mse.pdf** presents the LASSO cross-validation plot used during abundance estimation.

B. Intra-species diversity

Step 1: Intra-species reference database construction

Command	Description
#!/bin/bash module load strainest strainest mapgenomes SC*.fna Xp91-118.fna MR.fna	# Load the strainest module # # Map individual strain genomes (SC*.fna) to the reference genome located in the Inter_species folder.
strainest map2snp Xp91-118.fna MR.fna snp.dgrp	# Extract SNPs from the multiple alignment.
strainest snpdist snp.dgrp snp_dist.txt hist.pdf	# Calculate distances based on SNP profiles, outputting a distance matrix and histogram.

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strainest snpclust snp.dgrp snp_dist.txt snp_clust.dgrp clusters.txt	# Cluster the genomes based on SNP distances to group similarities.
strainest mapgenomes SC*.fna Xp91-118.fna MA.fna	# Generate a final representative multiple sequence alignment (MA.fna) for mapping. .
source /apps/profiles/modules_asax.sh.dyn module load bowtie2/2.5.1	# Setup environment for Bowtie2 (Note: redundant loading from the original script is kept for fidelity)
bowtie2-build -f MA.fna MA	# # Build a Bowtie2 index from the multiple alignment FASTA file (MA.fna). # '-f' specifies the input is a FASTA file. 'MA' is the base name for the output index files. # Build a Bowtie2 index from the multiple alignment FASTA file (MA.fna).

Step 2: Read Mapping and quantification of Intra-species diversity

Here, the step and command line are similar from the Inter-species diversity (A Step 2). We can use the same commands for this step as well.

Loop Script for multiple samples:

```
# Loop through all clean_trim1.fq files in the specified directory
for fq1 in ./*.paired.1.fq
do
    echo "Working with file: $fq1"

    # Extract base name without extension
    base=$(basename $fq1 .paired.1.fq)
    echo "Base name is: $base"

    # Define fq2 file path for paired-end reads
    fq2="./${base}.paired.2.fq"

    # Run Bowtie2 to align the reads
    bowtie2 --very-fast --no-unal -x /referenc/MA -1 $fq1 -2 $fq2 -S ./${base}.sam

    # Convert SAM to BAM, sort, and index the BAM file
    samtools view -b ./${base}.sam > ./${base}.bam
    samtools sort ./${base}.bam -o ./${base}.sorted.bam
    samtools index ./${base}.sorted.bam

    # Run strain estimation on the sorted BAM file
    strainest est snp_clust.dgrp ./${base}.sorted.bam ./${base}

done
```

8. RESULTS AND INTERPRETATION

We first evaluated the pipeline at the interspecies level using a metagenomic sample to detect the presence of four *Xanthomonas* species. The abundance estimates identified *X. perforans* strain **Xp91-118** as the dominant species, representing **94.97%** of the sample, while *X. euvesicatoria* strain **Xeu_85-10** accounted for the remaining **5.03%**. No abundance was detected for *X. gardneri* (**Xg_USA01-FM3**) or *X. vesicatoria* (**Xv_LMG911**), indicating that these species were absent from the sample (as shown in the snapshot of the result abund.txt file below). These results show that the pipeline can accurately distinguish species present in a mixed metagenomic sample.

OTU	sample.	sorted.bam
Xp91-118.fna	0.949714	
Xg_USA01-FM3.fna	0.000000	
Xeu_85-10.fna	0.050286	
Xv_LMG911.fna	0.000000	

After identifying *X. perforans* as the predominant species, we performed a second analysis to investigate intra-species diversity (strains/lineage identification) within this species. Using a reference database containing only *X. perforans* strains, the pipeline estimated that the artificial sample consisted primarily of strains **SC3_JK5-1 (56.54%)** and **SC1_DSM18975 (37.61%)**, with a smaller contribution from **SC6_Tu-09 (5.84%)** (as shown in the snapshot of the result abund.txt file below). All other *X. perforans* strains showed predicted abundances of **0%**, indicating that they were not present in the sample.

OTU	sample_intra.	sorted.bam
SC3_JK5-1.fna	0.565432	
SC5_AL33.fna	0.000000	
SC6_Tu-09.fna	0.058433	
SC8_JK42-6.fna	0.000000	
SC2_JK54-6.fna	0.000000	
SC7_BRIP62405.fna	0.000000	
SC1_DSM18975.fna	0.376135	
SC4_JK23-1.fna	0.000000	

Together, these results demonstrate a two-step approach for characterizing pathogen diversity. The pipeline first identifies the species present in a metagenomic sample and then resolves the relative abundance of individual strains within the selected species, enabling high-resolution analysis of pathogen populations.

9. SUGGESTIONS FOR DETERMINING STUDENT LEARNING

Power of this pipeline:

This configuration provides enough discriminatory power to cleanly separate highly similar (~99%), closely related clonal organisms or local multi-strain outbreaks, identifying variations with fine-grained precision. In complex diagnostic mixtures, a 99% threshold corresponds to distinguishing profiles with a minimum of 250 to 1,150 discordant SNV sites.

Limitations:

Researchers must keep these boundaries in mind during their experimental designs:

- **Known-lineage Bias:** If a field sample contains a wild/novel strain that is completely missing from the reference database, the pipeline may overestimate sample complexity by modeling it as a mixture of known reference strains, or it may fail to map divergent reads entirely.
- **Sequencing Depth Bias:** Distinguishing truly low-frequency strains in field samples requires high coverage depth. Low-coverage positions must be heavily filtered out (e.g., by discarding sites below defined cutoff thresholds), which limits fine-strain profiling in lower-yielding or highly contaminated environmental samples.
- **Recombination sensitivity:** Like all molecular typing schemas based on conserved marker loci, StrainEst relies on genetic linkage between these markers and features of interest. Some species are more recombinogenic than others, and if linkage is insufficient, more specialized, feature-specific methods are needed.
- **Strain misidentification:** For strains that are >99% identical with few SNPs, StrainEst may fail to reliably distinguish them. It could have limited statistical power to differentiate them in mixtures

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Data availability:

All the code files, including input and output files, are available on this GitHub repository: <https://github.com/Potnislab/APS-Plant-Health-Workshop-2026>.

References:

- Albanese, Davide, and Claudio Donati. 2017. "Strain Profiling and Epidemiology of Bacterial Species from Metagenomic Sequencing." *Nature Communications* 8 (1): 2260. <https://doi.org/10.1038/s41467-017-02209-5>.
- Bhandari, Rishi, Amanpreet Kaur, Ivory Russell, et al. 2025. "Interaction Between Climatic Variation and Pathogen Diversity Shape Endemic Disease Dynamics in the Agricultural Settings." *Molecular Ecology* 34 (17): e70033. <https://doi.org/10.1111/mec.70033>.
- Kim, Daehwan, Li Song, Florian P. Breitwieser, and Steven L. Salzberg. 2016. "Centrifuge: Rapid and Sensitive Classification of Metagenomic Sequences." *Genome Research* 26 (12): 1721–29. <https://doi.org/10.1101/gr.210641.116>.
- Langmead, Ben, and Steven L. Salzberg. 2012. "Fast Gapped-Read Alignment with Bowtie 2." *Nature Methods* 9 (4): 357–59. <https://doi.org/10.1038/nmeth.1923>.
- Li, Heng, Bob Handsaker, Alec Wysoker, et al. 2009. "The Sequence Alignment/Map Format and SAMtools." *Bioinformatics* 25 (16): 2078–79. <https://doi.org/10.1093/bioinformatics/btp352>.
- Lu, Jennifer, Florian P. Breitwieser, Peter Thielen, and Steven L. Salzberg. 2017. "Bracken: Estimating Species Abundance in Metagenomics Data." *PeerJ. Computer Science* 3: e104. <https://doi.org/10.7717/peerj-cs.104>.
- Nayfach, Stephen, Beltran Rodriguez-Mueller, Nandita Garud, and Katherine S. Pollard. 2016. "An Integrated Metagenomics Pipeline for Strain Profiling Reveals Novel Patterns of Bacterial Transmission and Biogeography." *Genome Research* 26 (11): 1612–25. <https://doi.org/10.1101/gr.201863.115>.
- Olm, Matthew R., Alexander Crits-Christoph, Keith Bouma-Gregson, Brian A. Firek, Michael J. Morowitz, and Jillian F. Banfield. 2021. "inStrain Profiles Population Microdiversity from Metagenomic Data and Sensitive Detects Shared Microbial Strains." *Nature Biotechnology* 39 (6): 727–36. <https://doi.org/10.1038/s41587-020-00797-0>.
- Scholz, Matthias, Doyle V. Ward, Edoardo Pasolli, et al. 2016. "Strain-Level Microbial Epidemiology and Population Genomics from Shotgun Metagenomics." *Nature Methods* 13 (5): 435–38. <https://doi.org/10.1038/nmeth.3802>.
- Truong, Duy Tin, Adrian Tett, Edoardo Pasolli, Curtis Huttenhower, and Nicola Segata. 2017. "Microbial Strain-Level Population Structure and Genetic Diversity from Metagenomes." *Genome Research* 27 (4): 626–38. <https://doi.org/10.1101/gr.216242.116>.
- Wood, Derrick E., Jennifer Lu, and Ben Langmead. 2019. "Improved Metagenomic Analysis with Kraken 2." *Genome Biology* 20 (1): 257. <https://doi.org/10.1186/s13059-019-1891-0>.
- Zolfo, Moreno, Adrian Tett, Olivier Jousson, Claudio Donati, and Nicola Segata. 2017. "MetaMLST: Multi-Locus Strain-Level Bacterial Typing from Metagenomic Samples." *Nucleic Acids Research* 45 (2): e7. <https://doi.org/10.1093/nar/gkw837>.